

VALTIMO KK, Finland

WMO: 027980

Lat: 63.667N

Lon: 28.829E

Elev: 377

StdP: 14.50

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.1	(c) -17.4	(d) -28.8	(e) 1.1	(f) -23.2	(g) -22.2	(h) 1.6	(i) -17.2	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	80.6	65.0	76.8	63.3	73.4	61.6	68.0	76.3	65.7	73.9	63.6	70.5	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.0	93.8	71.1	62.7	86.5	68.9	60.5	79.8	67.0	32.5	76.6	30.7	74.0	29.1	70.1	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -27.5	(f) 83.4	(g) 8.1	(h) 5.8	(i) -33.3	(j) 87.6	(k) -38.1	(l) 91.0	(m) -42.6	(n) 94.3	(o) -48.5	(p) 98.5		
(a) N/A	(b) N/A	(c) N/A	(e) -27.1	(f) 70.5	(g) 7.9	(h) 3.9	(i) -32.8	(j) 73.3	(k) -37.4	(l) 75.5	(m) -41.8	(n) 77.7	(o) -47.5	(p) 80.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.4	11.9	16.8	23.3	34.8	47.9	55.5	61.8	57.8	49.3	37.3	29.5	20.8
	DBStd	18.92	15.16	14.03	10.17	7.14	8.31	7.16	6.46	5.99	6.24	6.48	8.10	12.89
	HDD50	5572	1181	929	827	456	138	26	2	9	89	395	616	904
	HDD65	10157	1646	1349	1292	905	530	296	139	235	471	859	1066	1369
	CDD50	956	0	0	0	1	75	192	369	250	68	1	0	0
	CDD65	65	0	0	0	0	2	11	41	11	0	0	0	0
	CDH74	660	0	0	0	0	61	130	367	102	0	0	0	0
	CDH80	118	0	0	0	0	3	20	78	17	0	0	0	0
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	PrecAvg	25.9	1.9	1.5	1.3	1.3	2.0	2.6	3.4	2.8	2.3	2.4	2.3	2.2
(15) Precipitation	PrecMax	35.2	3.5	3.1	2.7	2.4	4.1	5.3	6.3	6.5	4.3	4.3	5.0	4.4
	PrecMin	17.4	0.5	0.2	0.2	0.4	0.5	0.8	1.3	0.7	0.5	0.5	0.5	0.7
	PrecStd	3.8	0.7	0.7	0.6	0.6	1.0	1.1	1.5	1.3	1.0	1.0	1.0	0.9
	PrecStd	3.8	0.7	0.7	0.6	0.6	1.0	1.1	1.5	1.3	1.0	1.0	1.0	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.8	40.1	46.4	63.7	79.7	82.7	85.9	82.1	69.0	54.3	47.3	39.6
		MCWB	34.6	37.0	38.9	49.3	58.6	64.7	69.4	67.8	60.1	51.4	45.5	38.7
	2%	DB	34.0	36.8	41.1	55.1	75.0	78.3	81.9	76.5	64.7	51.1	43.7	36.6
		MCWB	33.3	34.9	35.4	45.5	57.7	63.1	66.9	63.7	57.7	48.7	42.6	35.3
	5%	DB	32.5	34.5	37.9	50.6	69.2	73.1	77.9	72.5	61.4	48.8	41.0	34.9
		MCWB	32.0	33.5	34.3	41.8	55.4	60.4	65.3	62.1	56.2	46.8	40.4	33.9
	10%	DB	31.4	33.0	35.7	46.6	63.6	69.3	74.4	68.9	58.9	46.5	38.6	33.5
		MCWB	30.9	32.2	33.0	39.8	51.9	58.2	63.5	60.5	55.1	44.7	37.9	32.8
	0.4%	WB	35.4	37.6	39.4	50.5	61.4	67.4	71.7	70.0	61.3	52.2	46.2	39.2
		MCDB	35.9	39.6	44.1	60.0	74.3	75.4	80.4	80.2	65.9	53.4	47.3	39.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.4	35.3	36.7	46.6	59.1	65.0	69.2	65.7	59.3	49.5	42.8	35.7
		MCDB	33.7	36.5	39.9	53.4	70.6	74.9	77.8	73.9	63.0	50.6	43.6	36.4
	5%	WB	32.2	33.6	35.0	42.7	56.4	61.9	67.2	63.6	57.3	47.1	40.4	34.1
		MCDB	32.6	34.3	37.2	48.7	67.7	70.9	75.6	69.8	60.4	48.4	40.9	34.7
	10%	WB	30.8	32.4	33.6	40.4	53.0	59.3	65.2	61.8	55.5	44.8	38.0	33.0
		MCDB	31.2	33.1	35.3	45.2	62.5	66.9	72.4	67.3	58.3	46.4	38.5	33.5
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	12.7	16.6	16.7	18.6	17.0	16.6	15.7	12.8	8.8	6.7	9.1
		MCDBR	6.7	8.1	14.5	24.6	28.0	23.9	22.3	22.1	17.0	11.3	7.4	6.5
	5% WB	MCWBR	6.8	7.4	10.8	14.8	13.2	10.7	9.5	10.8	10.8	9.2	6.9	6.2
		MCDBR	6.4	8.0	11.7	21.1	25.3	20.0	19.8	18.1	14.9	10.6	7.4	6.1
	5% WB	MCWBR	6.5	7.5	9.4	13.3	12.9	9.8	9.8	10.0	10.1	9.5	7.0	6.1
		MCDBR	6.5	7.5	9.4	13.3	12.9	9.8	9.8	10.0	10.1	9.5	7.0	6.1
(40) Clear-Sky Solar Irradiance	taub		0.220	0.281	0.309	0.347	0.348	0.346	0.371	0.352	0.329	0.307	N/A	0.209
	taud		1.976	2.104	2.155	2.266	2.447	2.478	2.423	2.462	2.516	2.393	N/A	1.967
	Ebn at Noon		115	199	245	263	275	278	265	261	245	196	N/A	52
	Edh at Noon		10	21	30	33	31	30	31	28	22	16	N/A	4
(44) All-Sky Solar Radiation	RadAvg		54	227	689	1167	1489	1660	1565	1186	651	242	58	21
	RadStd		7	51	78	130	164	105	165	130	73	41	12	3

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	+0.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page